RANAJOY MUKHERJEE

TEST ENGINEER

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PROFILE

Detail-oriented Test Engineer with 3+ years of experience specializing in manual and API testing within Agile environments. Proven expertise in designing comprehensive test plans, executing test cases, and identifying defects to ensure high-quality software delivery. Skilled in functional, regression, and system testing with hands-on exposure to automation using Selenium. A collaborative team player proficient in defect tracking with JIRA and committed to upholding quality standards throughout the SDLC.

PROFESSIONAL EXPERIENCE

Test Engineer, Codelogicx Technologies, Kolkata **July 2024 — Present**

- Analyze software requirements and develop detailed test cases for comprehensive testing.
- Execute manual testing for functional, regression, integration, and system aspects of web applications.
- Conduct API testing with Postman to ensure backend functionality meets specifications.
- Manage defect lifecycle through JIRA, collaborating closely with developers for timely resolution.
- Participate in Agile/Scrum ceremonies to enhance team collaboration and project efficiency.

Software Engineer, Blue Copper Technologies, Kolkata **June 2023 — 2024**

- Developed and executed automated test scripts using Selenium with Java, driving a 30% increase in test coverage.
- Conducted API testing with Postman to validate backend functionality and maintain data integrity.
- Enhanced collaboration with cross-functional teams, ensuring adherence to quality standards across the SDLC.
- Implemented performance testing strategies utilizing JMeter for improved application reliability.
- Streamlined defect tracking and management processes using JIRA within Agile frameworks.

Test Engineer, Care Health Insurance **April 2022 — May 2023**

- Conducted end-to-end manual testing of health insurance application features.
- Performed API testing with Postman and basic performance testing with JMeter.
- Developed and executed test cases while creating traceability matrices for comprehensive coverage.
- Reported and tracked defects to resolution, ensuring a high-quality user experience.
- Collaborated with cross-functional teams to enhance testing processes and improve product quality.

EDUCATION

B.Tech in Computer Science and Engineering

Techno India, Salt Lake

Higher Secondary

Naihati Narendra Vidyaniketan

Secondary

Naihati Narendra Vidyaniketan

TECHNICAL SKILLS

Functional	Regression	System	Integration
End-to-End	UI	Usability	Compatibility
Sanity	Smoke	Postman	JMeter
Selenium WebDriver (Basic)	JIRA	Bugzilla	DBMS (MySQL, Oracle)
Agile	Scrum	STLC	Defect Life Cycle
Java (Basic)	SQL		

PROJECTS

Tractus – Health Insurance Product

Analyzed business requirements to develop and document test scenarios and cases for functional and regression testing. Executed test cases, collected test data, and reported defects, ensuring the quality of an integrated application for corporate and retail customers.

CanPay Debit – Payment Application

Developed test plans for integration testing of ACH transactions and validated end-to-end data integrity. Identified and reported defects in payment processing workflows and validated file processing via SFTP server uploads.

MDG Omnia - Shipment Tracking & Quality Assurance Platform

PROJECT:

Description: Led the development/implementation of MDG Omnia, a comprehensive based platform designed for end-to-end real-time shipment tracking and quality monitoring. The system integrates hardware sensors with a software analytics dashboard to ensure integrity and compliance of sensitive goods during transit.

Key Responsibilities & Contributions:

- Spearheaded the design of a real-time tracking module, providing live visibility into shipment location and status, reducing manual check inquiries by an estimated 40%.
- Engineered the integration with sensors to continuously capture and transmit critical environmental parameters including **temperature, humidity, pressure, shock, and tilt**.
- Developed the core excursion detection algorithm to automatically identify and flag when sensor values breached pre-defined client thresholds.
- Implemented a proactive alarm module that triggers immediate multi-channel alerts (email, SMS, dashboard) for critical excursions, enabling rapid corrective action and minimizing spoilage.
- Built analytics and reporting features to generate historical data logs, excursion reports, and compliance certificates for audits and process improvement.

Technical Environment: IoT Sensors, Cloud Architecture, Real-Time Data Processing, Predictive Analytics, Dashboard Development (e.g., Power BI/Tableau), Python/Node.js, SQL

Key Achievements:

- **Enhanced Compliance:** Directly ensured cold chain integrity for pharmaceutical and life sciences clients, guaranteeing adherence to strict regulatory standards (e.g., FDA, WHO).
- **Reduced Losses:** Minimized financial losses by enabling proactive intervention during transit excursions, preventing spoilage of high-value goods.
- **Improved Accountability:** Provided data-driven insights into carrier performance and handling conditions, strengthening vendor negotiation and contract management.
- **Automated Reporting:** Streamlined the audit process with automated, tamper-evident historical reports, saving countless manual hours.

Use Cases / Impact:

- **Pharma & Life Sciences:** Monitored cold chain shipments (2°C - 8°C), ensuring vaccine and medication efficacy.
- **Food & Beverages:** Protected perishable goods by monitoring freshness parameters, ensuring safety for end-consumers.
- **Electronics/High-Value Goods:** Detected and documented mishandling events (shock, tilt), providing evidence for insurance claims.
- **Chemicals:** Safely transported hazardous materials by ensuring stable environmental conditions throughout the logistics chain.